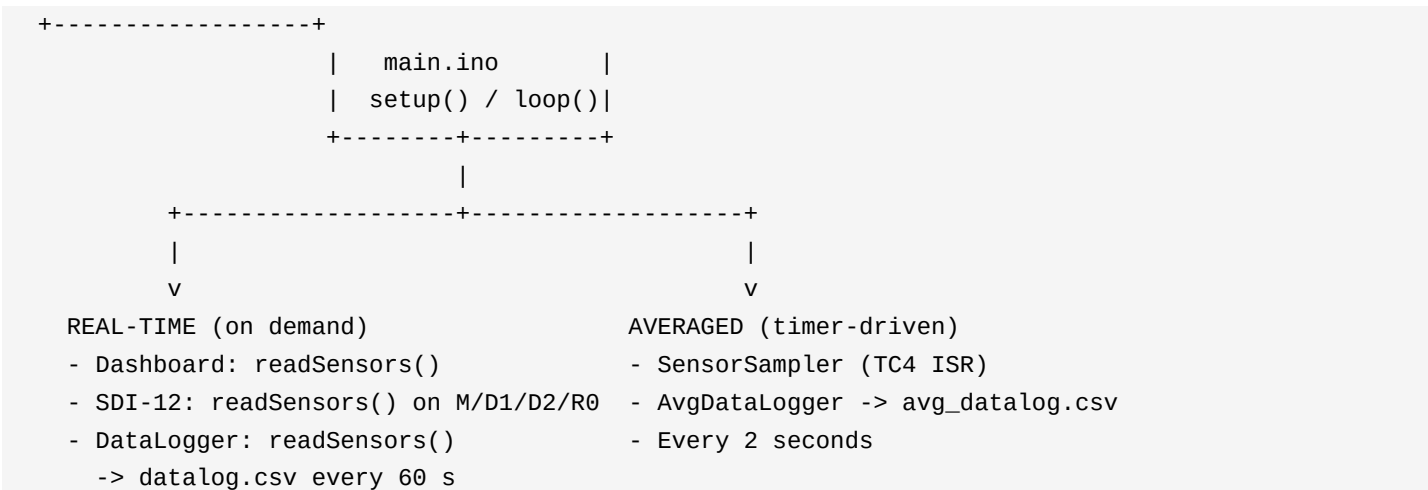


SDI-12 Sensor Station - System Workflow

Board: Arduino Zero / MKR / SAMD21 (TC4 hardware timer). This document explains how main.ino and all modules work together.

1. Big picture (two separate sensor paths)

- REAL-TIME path: instant I2C reads for TFT, SDI-12, and datalog.csv
- AVERAGED path: TC4 timer every 2 s -> avg datalog.csv only



2. Module map (what each file does)

File	Role
main.ino	Entry point. Calls all init and loop handlers in order.
HardwareConfig.h	Pins, intervals: dashboard 1s, datalog 60s, sample 2s.
SensorReading.cpp	BME280 + BH1750 I2C. readSensors(), getSensorData().
SensorSampler.cpp	TC4 ISR (10ms base, 2s sample). Builds 2s averages.
SdCard.cpp	One SD card + RTC shared by both loggers.
DataLogger.cpp	datalog.csv - periodic (60s) + manual button.
AvgDataLogger.cpp	avg_datalog.csv - one row per 2s average.
Dashboard.cpp	TFT display - real-time readSensors(), NOT averages.
sdi12.cpp	SDI-12 slave on Serial1. Uses getSensorData() after read.

3. setup() order (runs once at power-on)

```
Serial.begin(9600)
  sensorsInit()          // Wire + BME280 + BH1750
  sdCardInit()           // RTC + SD card (shared)
  sensorSamplerInit()    // Start TC4 timer (SAMD21)
  dataLoggerInit()       // Buttons + datalog.csv header
  avgDataLoggerInit()    // avg_datalog.csv header
  dashboardInit()        // TFT labels
  sdi12Init('0', pin 8) // SDI-12 bus
  readSensors()          // First reading
```

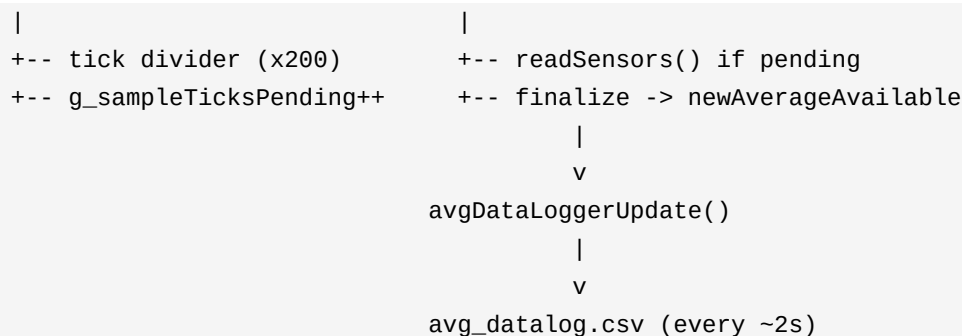
4. loop() order (runs forever)

- ```
1. sensorSamplerService() // FIRST: process 2s timer flags, I2C for averages
2. sdi12Handle() // SDI-12 commands + R0 stream (1s)
3. avgDataLoggerUpdate() // Write avg_datalog.csv when new 2s average ready
```

```
4. dataLoggerUpdate() // 60s log + manual/clear buttons
5. dashboardUpdate() // TFT refresh every 1s, real-time readSensors()
```

## 5. Averaged path detail (SensorSampler)

- TC4 hardware interrupt fires every ~10 ms (base tick).
- ISR counts 200 ticks -> one sample request every 2000 ms (2 s).
- ISR only sets g\_sampleTicksPending (NO I2C inside interrupt).
- sensorSamplerService() in loop reads sensors and accumulates.
- After 1 sample in window -> finalizeAverageWindow().
- AvgDataLogger writes row and calls sensorSamplerAcknowledgeAverage().



## 6. Real-time path detail (everything else)

- Dashboard: every 1000 ms calls readSensors() then getSensorData().
- DataLogger: every 60000 ms OR button pin 2 -> datalog.csv row.
- SDI-12: host sends command ending with ! -> readSensors() -> reply.
- SDI-12 R0 stream: every 1000 ms while streaming -> readSensors().
- Clear SD button (pin 3): deletes datalog.csv only (not avg file).

## 7. CSV files (do not mix them up)

| File            | Interval      | Data type                                |
|-----------------|---------------|------------------------------------------|
| datalog.csv     | 60 s + manual | Instant snapshot at log time             |
| avg_datalog.csv | ~2 s          | Hardware-timed average (1 sample/window) |

## 8. SDI-12 commands (address 0 by default)

- ? or 0! -> Respond with sensor address
- 0M! -> Measure: returns count of parameters
- 0D1! -> Send BME: temp, humidity, pressure
- 0D2! -> Send BH1750: lux
- 0R0! -> Send all + start continuous stream every 1 s
- 0An! -> Change address to n

## 9. Timing constants (HardwareConfig.h)

- kSensorSampleMs = 2000 ms (Average path: time between I2C samples)
- kSensorAverageMs = 2000 ms (Length of one average window)
- kTc4BasePeriodMs = 10 ms (TC4 hardware tick (divider -> 2 s))
- kDashboardRefreshMs = 1000 ms (TFT update rate)
- kLogIntervalMs = 60000 ms (datalog.csv auto-log interval)
- R0\_STREAM\_INTERVAL\_MS = 1000 ms (SDI-12 R0 stream (in sdi12.cpp))

## 10. Common confusion - quick answers

Q: Why two sensor reads?

TC4

A: San

**Q: Does TFT show averages?**

A: No.

**Q: Does SDI-12 use averages?**

A: No.

**Q: What does TC4 do?**

A: Onl

**Q: Non-SAMD board?**

A: Ser

*Generated for: Testing Environment/main/*